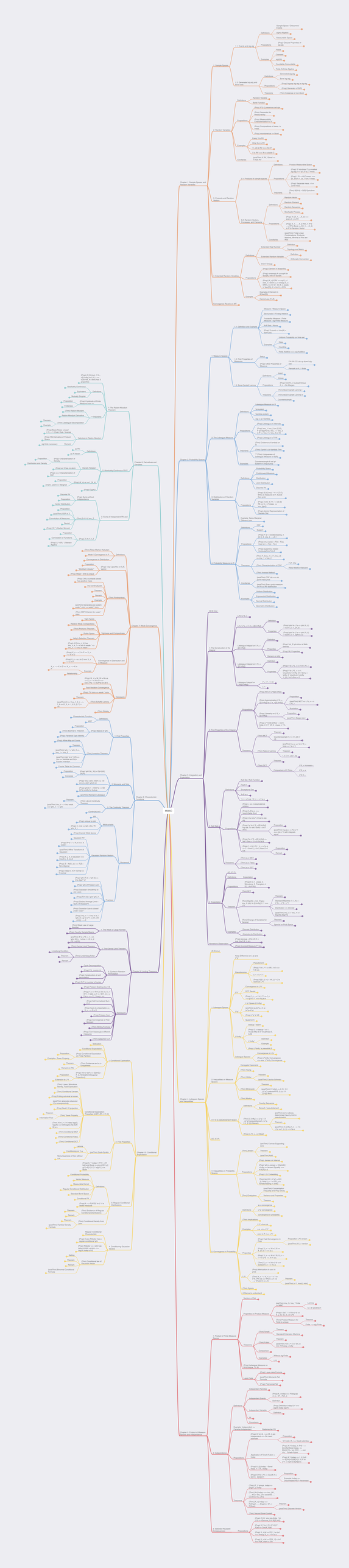

MMIO
Chapter 1: Sample Spaces and Random Variables
1. Sample Spaces
1.1. Events and σ-algebras
Definitions
Sample Space, Outcomes, Events
σ-algebra (𝒜)
Measurable Space
Propositions
(Prop) Closure Properties of σ-algebras
Proof
Examples
Coinflip
Dice
Countable Uncountable
Finite Cofinite Algebra
1.2. Generated σ-algebras and Borel sets
Definitions
Generated σ-algebra
Borel σ-algebra
Propositions
(Prop) Algebra that Fails to Be a σ-algebra
(Prop) Generation of 𝒢
Theorems
(Thm) Existence of non-Borel Sets
2. Random Variables
Definitions
Random Variable
Borel Function
Propositions
(Prop) σ(f) Lift Theorem
(Prop) Generator for Measurability
(Prop) Measurability Characterization for ℝ
(Prop) Compositions of Cont. is Meas.
(Prop) measurability on Borel
Examples
Every ℝ is RV
Only ∅ or Ω is RV
Constant RV
Indicator RV
Coinflips
Coinflips
Coinflips